

CENG 317 - Homework 3

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1. Performance Metrics

The experimental study was conducted by running both Hill Climbing and Simulated Annealing algorithms 10 times for each graph size (N = 10, 20, 50, 100). The "Success Rate" strictly measures the percentage of runs that reached **exactly 0 conflicts**.

Graph Size (N)	Algorithm	Success Rate (%)	Average Iterations	Final Conflicts (Avg. of Failed Runs)
N = 10	Hill Climbing	70.00%	100	1.33
	Simulated Annealing	10.00%	100	4.33
N = 20	Hill Climbing	50.00%	400	1.40
	Simulated Annealing	0.00%	400	5.20
N = 50	Hill Climbing	30.00%	2500	2.71
	Simulated Annealing	0.00%	2500	11.40
N = 100	Hill Climbing	10.00%	10000	3.33
	Simulated Annealing	0.00%	10000	20.40

2. Discussion: Hill Climbing vs. Simulated Annealing

Why Hill Climbing Gets "Stuck"

Hill Climbing is a purely greedy local search algorithm. At every step, it only makes moves that strictly improve or maintain the current state (i.e., reducing the number of conflicts). While this allows it to quickly find a good state, it often gets trapped in a **local optimum**. A local optimum occurs when the algorithm reaches a state where changing the color of any single node would *increase* the number of conflicts, but the total conflict count is still greater than zero. Because Hill Climbing refuses to make a move that increases conflicts, it has no way to escape this state and gets permanently stuck.

How Simulated Annealing Handles Local Optima

Simulated Annealing solves the local optimum problem by introducing a probability of accepting **worse states**. By utilizing a temperature parameter (T) and evaluating the energy difference (Delta E), the algorithm will occasionally accept a color change that increases the conflict count (with probability $P = e^{\{-\Delta E / T\}}$).

When the temperature is high (at the beginning of the search), it accepts worse states frequently, allowing it to freely explore the state space and "jump out" of local optima. As the temperature gradually cools down over time, the probability of accepting worse states decreases, and the algorithm begins to behave more like Hill Climbing, fine-tuning the solution to settle into a global optimum.
